

AI KNOWLEDGE ASSISTANT - SAMPLE COMPANY GUIDE

1. PROJECT OVERVIEW

The AI Knowledge Assistant is an internal application that helps employees

search company documents using natural language. Instead of manually opening many files, an employee can ask a question and receive an answer based on the approved company documents.

2. SUPPORTED DOCUMENTS

The prototype accepts PDF, DOCX, and TXT files. Documents are placed in a

local documents folder before ingestion.

3. RAG PIPELINE

The application follows four major stages:

Stage 1 - Document ingestion:

The application reads supported files and extracts their text.

Stage 2 - Chunking:

Large documents are divided into smaller text chunks. A chunk overlap is used so that information near chunk boundaries is less likely to be lost.

Stage 3 - Embeddings and vector storage:

Each chunk is converted into a numerical vector called an embedding. The embeddings and their associated text are stored in ChromaDB.

Stage 4 - Retrieval and generation:

When an employee asks a question, the question is converted into an embedding. ChromaDB searches for the most similar chunks. The retrieved chunks are supplied to the language model as context. The language model then generates an answer using that context.

4. TECHNOLOGY STACK

The prototype uses Python as the programming language.

LangChain is used to organize document loading, text splitting, retrieval,

prompts, and the LLM workflow.

PyPDF is used to extract text from PDF files.

ChromaDB is used as the local vector database.

Gemini is used for embeddings and for generating natural-language answers.

python-dotenv is used to load the API key from the .env file.

5. ANSWER POLICY

The assistant should answer using only information found in the retrieved

document context. If the answer is not available in the provided documents, the assistant should clearly say that it could not find the answer in the provided documents.

The assistant should not invent company policies, numbers, dates, names,

or technical details.

6. EXAMPLE

Question:

What is ChromaDB used for?

Answer:

ChromaDB is used as the local vector database where document chunks and their embeddings are stored so relevant information can be retrieved for a user's question.

7. SECURITY

API keys should be stored in environment variables and should not be

hard-coded into Python source files or committed to GitHub.

Internal documents should only be used in environments approved by the

organization. Access controls should be added before deploying the application for multiple employees.

8. FUTURE ENHANCEMENTS

Possible improvements include a Streamlit chat interface, drag-and-drop

file upload, chat history, source citations, metadata filtering, hybrid search, reranking, FastAPI, Docker, automated document ingestion, Azure AI Search, Azure OpenAI, authentication, and cloud deployment.

9. BUSINESS VALUE

A document RAG assistant can reduce the time employees spend searching

through large collections of policies, technical guides, procedures, and project documentation. It can also provide source references so employees can verify the information used to answer their questions.